GenAI for Engineers — Lesson Plan

IES | v4.0 | Aligned to supplied v2 PUB Feedback structure

Course duration: one day, 09:00–17:00. 390 teaching minutes, 60-minute lunch and two 15-minute tea breaks. Delivery is instructor-led and activity-based. No prior FMEA, DMAIC or SPC background is assumed.

Learning outcomes

LO	Outcome
LO1	Explain FMEA, Six Sigma/DMAIC and SPC before applying them.
LO2	Automate engineering documents and analyses with Copilot in Word, Excel and PowerPoint.
LO3	Apply R.I.S.E. and E-MAP with iterative prompting.
LO4	Build and deploy a multi-step Copilot Agent.
LO5	Use Researcher and Analyst for cited research and data analysis.
LO6	Ground an agent on SharePoint with cited, traceable answers.
LO7	Apply the complete 7-Step AI Security Framework.

Detailed lesson plan

Time	Module	PPT	Content and activity	LO
09:00–10:00	1 · Copilot ecosystem	Slides 5–25	Ecosystem, productivity, CW-401 case; access check; AI-vs-engineer challenge	LO2, LO3
10:00–11:15	2 · Prompting	Slides 26–55	RISE, E-MAP, constraints, iterative prompting, personas/debate; Word and PowerPoint document demos	LO2, LO3
11:15–11:30	Tea break	—	Break	—
11:30–13:00	3 · FMEA → DMAIC → SPC	Slides 60–98	Primer 3.0; Lab 3.1 FMEA; Lab 3.2 DMAIC; Lab 3.3 SPC	LO1, LO2, LO3
13:00–14:00	Lunch	—	Break	—
14:00–15:15	4 · Agents	Slides 95–124	FMEA Agent multi-step build/test; Report Writer/recurring	LO4, LO5

			workflow; Researcher and Analyst	
15:15–15:30	Tea break	—	Break	—
15:30–16:15	5 · SharePoint	Slides 125–143	Knowledge library; folder-scoped agent; citations, permissions and checked handoff	LO6
16:15–16:45	6 · AI security	Slides 144–156	Apply seven controls and consolidate personal toolkit	LO7
16:45–17:00	Closing	Slides 157–162	Outcome review, next action and Q&A	All

Module 3 methodology and lab allocation

Sequence	Concepts taught before activity	Guided activity	Evidence
FMEA	Purpose/use; function, mode, effect, cause, control; S/O/D; RPN; severity gate; action lifecycle	Lab 3.1 · 30 min	FM-01 = 140; FM-02 = 192; FM-04 escalates on Severity 9; ranked table and RPN chart
DMAIC	Six Sigma; five phases; Pareto; fishbone/Ishikawa; 5-Whys; facts separated from hypotheses	Lab 3.2 · 30 min	36/400 = 9.00% vs 7/200 = 3.50%; change stated as 5.50 percentage points; Pareto chart
SPC	Chart anatomy; common vs special cause; limits from a frozen baseline; control vs specification limits; Cp and Cpk	Lab 3.3 · 30 min	Subgroup 13 Xbar = 61.275 below LCL 61.547; Cp >> Cpk read as off centre, not incapable

The classroom-case order is fixed: FMEA → DMAIC → SPC. These methods answer different questions and are not a universal lifecycle. Every concept is explained before the corresponding Copilot application. Module 3 stops at 13:00.

Copilot integration across the day

Module	Copilot integration	Human verification
1	Chat structures the pump brief; app selection	Facts, assumptions and evidence boundary
2	Word/PowerPoint document drafts using RISE/E-MAP	Requirements, criteria, units and approval
3	Excel calculations, tables and charts	RPN, rates, control limits and interpretation
4	FMEA Agent workflow; Researcher; Analyst	Negative tests, citations and four anchor values
5	SharePoint-grounded agent and cross-agent handoff	Source passage, revision, permissions and audience
6	Security evidence log and release decision	All seven controls and named sign-off

Assessment of learning

LO	Formative evidence
LO1	Correctly explains method purpose, terms and output.
LO2	Produces checked FMEA, DMAIC and SPC outputs and critiques Word/PPT drafts.
LO3	Labels RISE/E-MAP content and improves one prompt after checking.
LO4	Agent passes complete-row and missing-data tests.
LO5	Researcher citation opens; Analyst calculations reproduce.

LO6	Agent gives cited supported answer and preserves missing evidence.
LO7	Seven completed evidence rows and approve/revise/reject decision.

Resources

- PPT/PDF: 186 slides; slides 1–162 support the timed route, 163–186 are optional quick references.
- Learner Guide: detailed concepts, procedures, prompts, checks and Microsoft links.
- Facilitator Guide: timing, teaching points, expected outputs, misconceptions and fallbacks.
- Sixteen Lesson Plan activity folders: each has a PDF, prompt, synthetic mock data and acceptance checks; Labs 3.1 and 3.2 also contain Excel workbooks.
- Trainer demonstration pack: agent, reasoning-agent, SharePoint and security resources.

Quality and contingency rules

- Module divider slides are full-page separators and contain no learning-content cards.
- Use visuals to explain mechanisms; use the Learner Guide for click-by-click procedures.
- Keep all activities inside their published module time boxes; use reference slides selectively.
- If live Copilot access fails, use the worked reference and complete the same independent check.
- All case data are synthetic. IES is the only provider branding.